

Historical Evidence for Off-Policy Evaluation of ETF Allocation Policies

When Better Value Estimates Do Not Resolve Policy Choice

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Abstract

Historical off-policy evaluation (OPE) is useful for deployment only when the available evidence can resolve a policy decision, not merely estimate policy values with moderate accuracy. We study this distinction with QQQ, SPY, and IWM in a one-step contextual-bandit design. OPE receives partial-feedback logs generated by a common Thompson-sampling logger, while the price-taking structure allows all three one-step action rewards to be reconstructed after the next return is observed and withheld for scoring. Under a frozen cross-ETF protocol, all available pre-anchor history gives lower doubly robust (DR) mean absolute error than trailing 252-, 126-, and 60-observation windows for every ETF. Yet even under this best-performing evidence rule, median candidate–baseline gaps are only 0.26–0.35 basis points versus paired bootstrap standard errors of 1.86–3.42 basis points; point-estimate policy selection is near 50%, and an approximate 95% one-sided lower-confidence rule admits the candidate in only about 2–3% of evaluation windows. Improving target-policy action coverage raises effective sample size by 3.33–3.84% at $m = 80$, but lowers MAE by only 0.71–1.86%. The central finding is therefore decision-based: more history and better overlap can improve standard OPE diagnostics without making a close policy comparison resolvable. Historical OPE should distinguish value accuracy, action support, and policy-gap resolution, and should permit evidence-based abstention when the logged data do not justify a switch.

Introduction

Historical market data are often the only evidence available for deciding whether a new allocation policy should replace an existing one. The practical question is therefore not only whether off-policy evaluation (OPE) can estimate each policy’s value, but whether the historical evidence is precise and relevant enough to resolve the policy choice. A value estimate can look reasonable while the estimated difference between two close policies remains too uncertain to support deployment.

OPE estimates target-policy value from data generated by another behavior policy (Dudík, Langford, et al. 2011; Jiang and L. Li 2016; Wang et al. 2017). In historical financial data, three distinct features can limit decision usefulness. Older observations enlarge the evidence base but may be less representative of current conditions. Target-policy actions may have weak support under the behavior policy. Finally, the economically relevant difference between a candidate and a baseline may be smaller than the uncertainty of their estimated value difference. These are different problems: a recent sample can still have weak overlap, high effective sample size does not guarantee temporal relevance, and low marginal value error does not imply that two close policies can be ranked reliably.

We study these distinctions with QQQ, SPY, and IWM. Historical prices define the market paths, and a stochastic Thompson-sampling logger creates partial-feedback behavior data. At each decision time the action reduces, keeps, or increases ETF exposure by ten percentage points. Under the price-taking assumption, once the next return is observed the one-step reward of every action can be reconstructed at the same pre-action exposure. Learning and OPE receive only the logged-action reward; the full one-step action-reward vector is withheld until scoring. The design therefore gives a truth-known benchmark for asking whether historical partial-feedback evidence would have supported the correct local policy comparison.

The central empirical finding is a separation between *value estimation* and *policy resolution*. Under the frozen cross-ETF protocol, all available pre-anchor history gives the lowest DR mean absolute error among the pre-specified evidence windows for QQQ, SPY, and IWM. Nevertheless, under this best-performing evidence rule the median candidate–baseline gap is only 0.26–0.35 basis points, compared with paired bootstrap standard errors of 1.86–3.42 basis points. Point-estimate policy selection is consequently near 50%. Improving target-policy action coverage raises effective sample size, but its effect on MAE is much smaller. Thus more history and better overlap improve useful diagnostics without, by themselves, resolving the close deployment decision studied here.

The contribution is an empirical decision-audit design rather than a new OPE estimator or a trading strategy. It keeps policy construction fixed, varies the historical evidence supplied to OPE, perturbs target-policy action coverage on the same recent contexts, and evaluates the candidate–baseline difference with paired uncertainty against a withheld one-step benchmark. The organizing question is simple: *does the available historical evidence justify switching policies, or should the evaluator abstain?* This framing turns a collection of OPE diagnostics into a decision consequence and makes explicit when improved estimation is still insufficient for policy selection.

An earlier single-ETF pilot study examined historical memory, action coverage, and policy uncertainty using QQQ (S. Li 2026). The present study substantially revises the computational implementation and evaluation protocol rather than simply adding two assets to that pilot: it freezes one cross-ETF design for QQQ, SPY, and IWM, expands the repeated behavior-log analysis, imposes time-ordered nuisance estimation, adds financial robustness checks, and makes candidate–baseline policy resolution the primary empirical target. We therefore treat the earlier paper as exploratory pilot evidence and the present paper as a systematic cross-ETF decision audit.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 defines the ETF decision problem, the withheld benchmark, and the policy-resolution target. Section 4 describes the logging and OPE protocol. Section 5 reports the empirical results, beginning with the policy-resolution finding. Section 6 gives robustness checks, and Section 7 discusses interpretation and limitations.

Related Literature

Off-policy evaluation estimates target-policy value from logged actions and their propensities. Importance sampling, self-normalization, and doubly robust estimators use this information in different ways (Dudík, Langford, et al. 2011; Swaminathan and Joachims 2015; Wang et al. 2017). Related doubly robust methods also appear in sequential and nuisance-estimation settings (Jiang and L. Li 2016; Kallus and Uehara 2020; Kallus, Mao, et al. 2022). We use these methods as established evaluation tools. The paper’s target is not estimator novelty, but whether the resulting evidence is sufficient to resolve a fixed policy comparison.

A separate literature studies changing reward processes and the reuse of historical data under nonstationarity (Besbes et al. 2014; Faury et al. 2021; Dudík, Erhan, et al. 2012; Hong et al. 2021; Liu et al. 2023). This work motivates treating temporal relevance as an empirical question rather than assuming that either a longer history or a more recent window must be preferable. Policy-improvement methods address a different but closely related issue: whether an estimated advantage is sufficiently well determined to justify changing policy (Thomas et al. 2015). That perspective is central here. We evaluate the paired candidate–baseline difference directly and treat failure to resolve its sign as an informative outcome rather than forcing a ranking from point estimates.

Recent work makes policy selection or operational readiness, rather than value estimation alone, the explicit target. Ryu et al. (2025) develop variance-adaptive confidence bounds for off-policy selection in contextual bandits. In marketplace settings, Shekhar and Howard (2026b) map logged evidence into certified, dominated, and unresolved policy candidates, while Shekhar and Howard (2026a) use a support-aware decision-support pipeline to distinguish direct launch from further validation. These papers are methodologically adjacent to the present study, but their contribution is new selection, certification, or decision-support machinery. Our contribution is different and deliberately empirical: using a frozen historical-finance benchmark with withheld one-step action rewards, we ask whether improvements in conventional OPE diagnostics actually make a fixed candidate–baseline decision resolvable. No new confidence bound or certification theorem is claimed here.

Finance provides a useful stress setting because historical evidence is finite, market conditions evolve, and backtest selection can create misleading confidence (Arnott et al. 2019; López de Prado 2019). Recent work in *The Journal of Financial Data Science* applies reinforcement learning to trading and portfolio problems (Zhang et al. 2020; Han 2023; Ayala et al. 2025), and Borkar and Jadhav (2024) survey the broader stock-trading literature. The present study is complementary: it does not propose or optimize a new trading rule. ETF allocation is used as a transparent environment in which the one-step action rewards can be reconstructed after the next market return is observed, making the quality of historical OPE auditable against a withheld benchmark.

These strands correspond to three different questions: how much historical evidence to retain, whether target actions are adequately represented in the log, and whether the remaining uncertainty is small enough to distinguish the policies that matter for deployment. Our empirical design places the three questions in one frozen cross-ETF protocol. Its main distinction is between *estimability of policy value* and *resolvability of policy choice*: a procedure can improve MAE or effective sample size and still leave the candidate–baseline decision unsupported by the data. Relative to the earlier QQQ pilot (S. Li 2026), the present paper uses a substantially revised implementation and common cross-ETF protocol, more repeated behavior logs, time-ordered nuisance estimation, robustness checks, and a decision-first ordering of the empirical analysis.

ETF Allocation Setting

ETF sample and common design

The empirical study uses three equity ETFs,

$$\mathcal{J} = \{\text{QQQ}, \text{SPY}, \text{IWM}\}. \quad (1)$$

The same features, action space, policy family, behavior logger, OPE methods, evidence windows, and inference rules are used for every ETF. The reconstructed one-step full-action benchmark is used only for scoring.

ETF	Final sample start	Final sample end	Decision observations
QQQ	2000-02-01	2026-08-27	6,683
SPY	2000-02-01	2026-08-27	6,683
IWM	2000-06-26	2026-08-27	6,582

Table 1: Final ETF samples after the common cleaning and feature-construction pipeline.

Prices were downloaded on August 29, 2026 from Yahoo Finance through `yfinance` 1.7.0 with automatic adjustment disabled, and the adjusted close was used consistently for all three ETFs. The common acquisition window begins on January 1, 2000 to keep the cross-ETF design comparable; the later final start dates in Table 1 reflect ETF availability and rows removed for rolling-feature construction. The feature pipeline also removes the final row without a next-period return. The raw and processed files, download timestamps, package versions, and SHA-256 hashes are stored with the reproducibility metadata. The next-period return is not included among the model features.

Context, action, and exposure

For ETF $j \in \mathcal{J}$, the context at decision time t is

$$x_{j,t} = \left[1, R_{j,t}^{(1)}, R_{j,t}^{(5)}, R_{j,t}^{(20)}, \sigma_{j,t}^{(20)}, M_{j,t}, D_{j,t}, w_{j,t} \right]^\top, \quad (2)$$

where the variables are short-horizon returns, twenty-day realized volatility, a moving-average spread, drawdown, and the pre-action ETF exposure. The next-period return is excluded from the context.

The action set is

$$\mathcal{A} = \{-1, 0, +1\}, \quad (3)$$

where -1 , 0 , and $+1$ mean reduce, keep, or increase exposure. With a fixed step $\Delta = 0.10$,

$$w_{j,t}^+(a) = \text{clip}(w_{j,t} + \Delta a, 0, 1). \quad (4)$$

One action therefore changes exposure by ten percentage points.

Reward and withheld one-step benchmark

The baseline reward is

$$r_{j,t+1}(a) = w_{j,t}^+(a)R_{j,t+1} - c \left| w_{j,t}^+(a) - w_{j,t} \right|, \quad c = 0.0005. \quad (5)$$

The reward isolates the one-step evaluation problem from additional strategy-engineering choices.

We assume that the simulated ETF position is price-taking,

$$R_{j,t+1}(a) = R_{j,t+1}, \quad a \in \mathcal{A}. \quad (6)$$

Given the logged pre-action exposure $w_{j,t}$, $w_{j,t}^+(a)$ is deterministic. Once $R_{j,t+1}$ is observed, Eq. (5) therefore gives the one-step reward for every action:

$$\mathbf{r}_{j,t+1}^{\text{cf}} = (r_{j,t+1}(-1), r_{j,t+1}(0), r_{j,t+1}(+1)). \quad (7)$$

Learning and OPE receive only the selected-action reward. The full vector is withheld until scoring. The term “counterfactual” is used here only in this one-step price-taking sense: the benchmark reconstructs alternative one-step rewards at the logged pre-action position and does not construct an alternative multi-period portfolio path.

For an evaluation anchor t and a forward window of H trading observations, the scoring target for policy π is

$$V_{j,t,H}^{\text{cf}}(\pi) = \frac{1}{H} \sum_{s=t}^{t+H-1} \sum_{a \in \mathcal{A}} \pi(a | x_{j,s}) r_{j,s+1}^{\text{cf}}(a). \quad (8)$$

Historical OPE estimates are scored by how closely they recover this realized finite-window one-step value.

Baseline and candidate target policies

For every ETF, action-specific Ridge reward models define two stochastic target policies. The baseline policy is trained with the most recent 756 trading observations before an evaluation anchor, and the candidate uses the most recent 126 observations. We refer to these as the *policy-training windows*.

The memory experiment changes a different quantity: the historical observations supplied to OPE. We call these the *OPE evidence windows*. The 756-versus-126 policy-training contrast is held fixed while the OPE evidence window is varied. This keeps policy construction separate from the question of how much historical evidence should be used to evaluate the policies.

Predicted action rewards are mapped through the same scaled-softmax construction for both target policies. The final protocol uses a common temperature of 1.50 and a common target-action probability floor of 0.02; the action-score scaling itself is estimated from each policy’s training sample with a fixed lower scale bound. Because the target policies are fitted from pre-anchor logged history that can also contribute to the OPE evidence, the evaluated policies are data-adaptive rather than independently trained on a separate sample. The forward scoring window is never used for target-policy fitting. The 756-versus-126 contrast is fixed for the main analysis, with a single 756-versus-252 contrast reserved for robustness.

Policy resolution as the decision target

For ETF j and evaluation anchor t , define the benchmark candidate–baseline gap

$$\Delta V_{j,t,H} = V_{j,t,H}^{\text{cf}}(\pi_{\text{new}}) - V_{j,t,H}^{\text{cf}}(\pi_{\text{base}}), \quad (9)$$

and its OPE estimate

$$\widehat{\Delta V}_{j,t,H} = \widehat{V}_{\text{DR}}^{(j)}(\pi_{\text{new}}) - \widehat{V}_{\text{DR}}^{(j)}(\pi_{\text{base}}). \quad (10)$$

The decision question is whether the logged evidence can recover the sign of ΔV with enough precision to justify replacing the baseline. Because the two policy values are estimated from the same log, uncertainty is computed for the paired difference rather than by combining two independent value intervals.

For evaluation only, we summarize the scale of the true policy difference relative to estimation uncertainty by

$$\mathcal{R}_{j,t,H} = \frac{|\Delta V_{j,t,H}|}{\widehat{\text{SE}}(\widehat{\Delta V}_{j,t,H})}. \quad (11)$$

This benchmark resolution ratio is not available to a real deployment system because its numerator uses the withheld full-action score. It is a diagnostic of the evidence regime: values far below one mean that the policy difference to be resolved is small relative to the uncertainty of the estimator. The operational one-sided rule instead uses only the bootstrap distribution of $\widehat{\Delta V}$ and admits the candidate only when its lower confidence bound is positive; otherwise it retains the baseline. In this paper, abstention is therefore an explicit decision outcome rather than an estimation failure.

OPE Protocol

Simulated behavior logs

ETF prices provide full market returns, while OPE is evaluated under partial feedback. For each historical ETF path, we simulate a behavior log with the same stochastic linear Thompson-sampling logger (Agrawal and Goyal 2013). Only the reward of the sampled action updates the logger, and its known action propensity is stored for OPE. Thompson sampling is used only to generate the partial-feedback logs. The logger uses 400 Monte Carlo draws per decision and a common action-probability floor of 0.03. Exact logger details are given in Appendix A.

OPE estimators and overlap

We compare Direct Method (DM), inverse propensity scoring (IPS), self-normalized IPS (SNIPS), and doubly robust (DR) evaluation (Dudík, Langford, et al. 2011; Jiang and L. Li 2016; Wang et al. 2017). For target policy π ,

$$\rho_i(\pi) = \frac{\pi(a_i | x_i)}{\mu_i(a_i | x_i)} \quad (12)$$

is the target-behavior importance ratio. With evidence weights ω_i , the main DR estimate is

$$\widehat{V}_{\text{DR}}(\pi) = \frac{\sum_i \omega_i \{ \widehat{m}_i(\pi) + \rho_i(\pi) [r_i - \widehat{q}(x_i, a_i)] \}}{\sum_i \omega_i}, \quad (13)$$

where $\widehat{m}_i(\pi) = \sum_a \pi(a | x_i) \widehat{q}(x_i, a)$. The other standard formulas are given in Appendix A.

Overlap is summarized by

$$\text{ESS}(\pi) = \frac{(\sum_i c_i)^2}{\sum_i c_i^2}, \quad c_i = \omega_i \rho_i(\pi). \quad (14)$$

ESS measures concentration in target–behavior weights. It is an evidence-quality diagnostic, not a policy-decision criterion. Historical relevance is a different issue: it concerns whether the reward relationship $P(r \mid x, a)$ has changed over time. Policy resolution is different again: it asks whether the uncertainty of the paired candidate–baseline contrast is small relative to the policy gap. Keeping these quantities separate is part of the empirical design.

OPE evidence windows

A shorter evidence window may use more current observations, but it also removes data and may weaken action support. The main analysis compares all observations available before the anchor with fixed trailing windows of 252, 126, and 60 trading observations. These choices were fixed before benchmark scoring and are used unchanged for all ETFs. A pre-specified exponential-decay sensitivity with a 126-observation half-life is retained only as supplementary output and is not used to choose the main evidence rule.

Time-ordered nuisance estimation

The final DR results use rolling-origin reward-model predictions. After a 252-observation warm-up, predictions are generated in blocks of 63 trading observations; the reward model for each block is trained only on observations strictly before that block. The implementation records the training-end date for each prediction and enforces `training_end < prediction_date`, avoiding future leakage and direct in-sample nuisance prediction.

Walk-forward scoring and benchmark use

The forward scoring horizon is fixed at $H = 80$ trading observations. The main analysis uses non-overlapping forward windows with anchor spacing of 80 observations. All data used to construct target policies, OPE evidence, and nuisance models occur before the scored window.

Each forward window is scored against Eq. (8). The withheld full-action benchmark is used only after the protocol is fixed and does not determine policy-training windows, OPE evidence windows, policy temperature, nuisance-model settings, or the selection rule.

Repeated logs, uncertainty, and decision metrics

The final experiment uses 20 independently simulated behavior logs per ETF. Standard estimation diagnostics include MAE, RMSE, signed bias, and ESS. The paper’s primary decision diagnostics are the candidate–baseline gap, its paired uncertainty, sign-based policy-selection accuracy, the rate at which point-estimate adoption chooses a truly worse candidate, and the frequency with which a one-sided lower-confidence rule has enough evidence to admit the candidate.

For paired policy-gap uncertainty, the pre-specified moving-block bootstrap uses 500 resamples and a block length of 20 observations. We use its empirical resampling distribution to form an approximate bootstrap-based 95% one-sided lower-confidence rule. Because this rule is used here as an approximate decision diagnostic and the paper does not establish a finite-sample coverage theorem for the moving-block bootstrap under the full data-adaptive protocol, the nominal 95% level should not be interpreted as a certified 95% deployment-safety guarantee. The same inference settings are used for QQQ, SPY, and IWM. Value errors and policy gaps are also reported in basis points, where 10^{-4} in one-step value equals one basis point.

Empirical Results

We begin with the policy-resolution result because it is the main deployment question. We then ask whether using more historical evidence or improving target-policy action coverage addresses that bottleneck, and finally report the standard estimator benchmark as a secondary diagnostic.

The historical evidence does not resolve the close policy choice

Table 2 compares the magnitude of the withheld candidate–baseline policy gap with the uncertainty of the DR estimate. The scale separation is large and consistent across all three ETFs.

ETF	Median $ \Delta V $ (bp)	Median SE (bp)	Gap/SE	Selection acc.	Worse-cand. rate
QQQ	0.31	3.42	0.09	50.1%	51.4%
SPY	0.26	1.87	0.14	50.2%	50.4%
IWM	0.35	1.86	0.19	48.8%	50.1%

Table 2: Candidate–baseline policy gaps and their estimation uncertainty. “Selection acc.” uses the sign of the point-estimated DR gap. The final column is the fraction of candidate adoptions for which the withheld benchmark shows that the candidate was actually worse.

Median absolute gaps are only 0.26–0.35 basis points, whereas median paired bootstrap standard errors are 1.86–3.42 basis points. The benchmark resolution ratio in Eq. (11) is therefore only 0.09–0.19: typical uncertainty is roughly five to eleven times the policy difference that must be resolved. Point-estimate selection is correspondingly near 50% for every ETF, and the approximate bootstrap-based 95% one-sided lower-confidence rule admits the candidate in only about 2–3% of evaluation windows. The low admission frequency is evidence that the rule usually finds the available historical log insufficient to justify a switch. It should not be interpreted as an empirical estimate or proof of a 95% safety rate; the nominal level belongs to the approximate bootstrap diagnostic described in Section 4, for which this paper does not claim a finite-sample deployment-safety guarantee.

This is the central empirical result of the paper. The near-50% selection rate should not be interpreted as evidence that the policies are identical. Rather, the historical partial-feedback evidence is not precise enough to recover the sign of a policy difference of this magnitude with useful reliability. In this regime, a forced point-estimate ranking behaves like an unstable decision rule, whereas a confidence-based procedure mostly abstains.

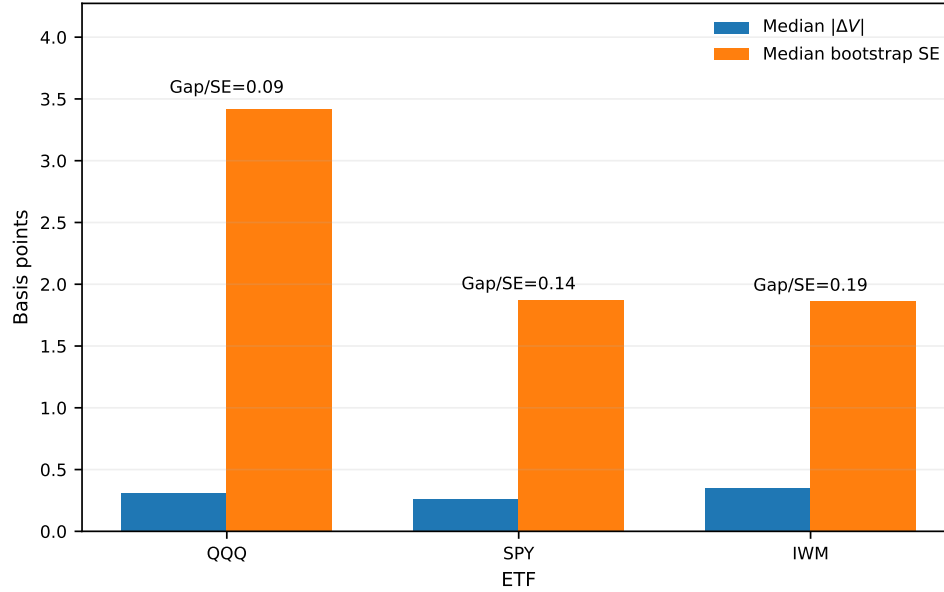


Figure 1: Median absolute candidate–baseline value gap and median paired bootstrap standard error. The gap-to-uncertainty ratio is far below one for QQQ, SPY, and IWM.

More historical evidence improves value accuracy, but not enough to resolve the decision

Holding candidate-policy construction fixed, we vary only the historical evidence supplied to DR. Table 3 reports the resulting MAE.

ETF	All history MAE (bp)	Last 252 MAE (bp)	Last 126 MAE (bp)	Last 60 MAE (bp)
QQQ	6.58	9.14	13.70	15.89
SPY	4.26	6.91	10.18	11.48
IWM	4.60	9.49	14.96	16.77

Table 3: DR OPE evidence-window comparison using the same pre-specified windows for every ETF.

All available pre-anchor evidence gives the lowest MAE for every ETF. Relative to all history, the 252-observation window raises MAE by 38.8% for QQQ, 62.1% for SPY, and 106.1% for IWM; the 126- and 60-observation windows increase error further. The uniform ordering does not establish a universal preference for all available history. Within the pre-specified one-step design, however, truncation sacrifices more information than it gains from temporal proximity.

The important decision implication is that the best tested evidence window is still not enough. The policy-resolution result above is measured under the full-history rule that has the lowest DR MAE. Thus improving value accuracy by retaining more data does not automatically cross the much harder threshold required to distinguish two policies whose values are separated by only a few tenths of a basis point.

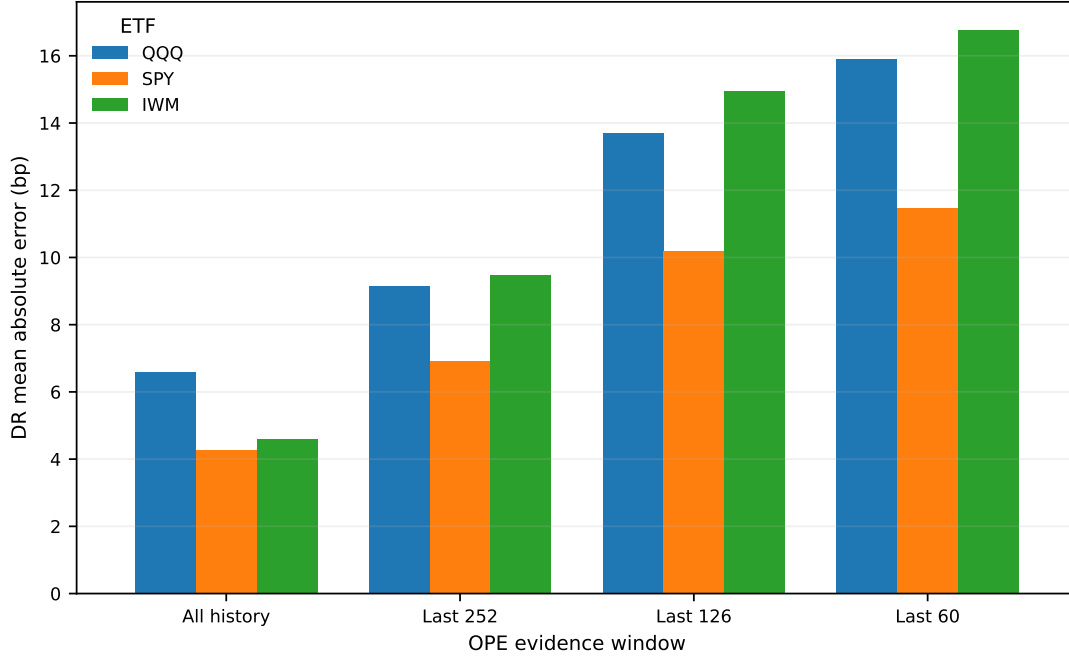


Figure 2: DR mean absolute error under the four pre-specified OPE evidence windows. All-history evidence has the lowest MAE for QQQ, SPY, and IWM, but the candidate–baseline decision remains unresolved under that best-performing rule.

Better action coverage improves ESS without a comparable gain in value accuracy

This experiment retains the historical evidence and adds observations from the same recent date pool. The recent-original condition uses the actions originally logged on those dates. The target-aware condition re-logs the same stored contexts from a behavior distribution equal to 0.8 times the baseline target policy plus 0.2 times the candidate policy; only the one-step reward associated with the newly sampled action is supplied to OPE. Thus the comparison changes action support without constructing an alternative multi-period market path.

ETF	Evidence source	m	MAE change	ESS change
QQQ	Recent original-policy	80	-2.38%	+2.79%
QQQ	Target-aware re-logging	80	-1.86%	+3.84%
SPY	Recent original-policy	80	-0.09%	+2.50%
SPY	Target-aware re-logging	80	-0.71%	+3.52%
IWM	Recent original-policy	80	-0.90%	+2.38%
IWM	Target-aware re-logging	80	-1.02%	+3.33%

Table 4: Recency and target-policy coverage on the same recent dates. Changes are relative to the history-only condition at the same evaluation anchors; negative MAE changes indicate improvement.

At $m = 80$, recent original-policy observations lower MAE by 0.09–2.38% and raise ESS by 2.38–2.79%. Target-aware re-logging produces larger ESS gains, 3.33–3.84%, but lowers MAE by only 0.71–1.86%. The full curves are small and not uniformly monotone; for SPY, the recent-original condition is essentially flat around $m = 40$. The result is deliberately narrow: better target–behavior overlap is visible in ESS, but the gain

does not translate proportionally into lower value-estimation error. ESS should therefore not be interpreted as a stand-alone measure of whether a close policy decision is resolvable.

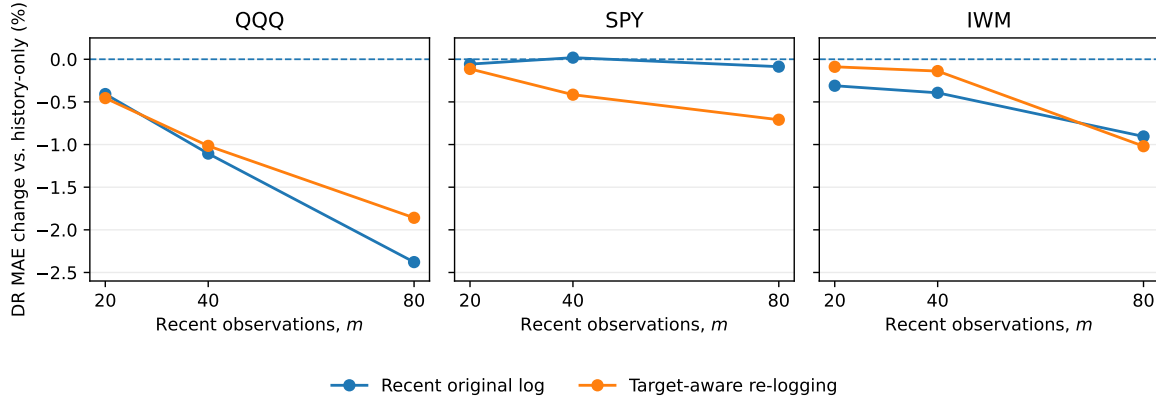


Figure 3: Percentage change in DR MAE relative to the history-only condition as recent observations are added. All three panels use the same vertical scale. The target-aware condition uses the same recent contexts while shifting action coverage toward the candidate policy.

Estimator choice does not yield a uniformly dominant benchmark

Table 5 documents the standard estimators under the common logging and scoring design. No estimator dominates across all three ETF paths: SNIPS has the lowest MAE for QQQ and SPY, whereas DM has the lowest MAE for IWM. DR is not the lowest-MAE estimator in this benchmark. Because DR was pre-specified for the evidence-window, coverage, and policy-resolution analyses, we retain it rather than selecting an estimator after observing the benchmark.

ETF	Estimator	MAE (bp)	Bias (bp)	Best-policy acc.
QQQ	IPS	6.65	-0.03	39.5%
QQQ	SNIPS	6.42	0.01	39.6%
QQQ	DR	7.90	-0.00	35.8%
QQQ	DM	8.55	-7.26	30.2%
SPY	IPS	5.02	-0.73	36.9%
SPY	SNIPS	4.87	-0.74	37.0%
SPY	DR	5.78	-0.84	36.6%
SPY	DM	6.09	-4.11	32.4%
IWM	IPS	6.79	0.35	40.2%
IWM	SNIPS	6.69	0.32	39.8%
IWM	DR	7.24	0.39	39.5%
IWM	DM	5.90	-2.46	33.8%

Table 5: Cross-ETF OPE benchmark under the frozen protocol. Best-policy accuracy is computed across the baseline, candidate, conservative-mixture, and uniform target policies within each evaluation block. Mean ESS is 33.8 for QQQ, 39.9 for SPY, and 40.9 for IWM for every estimator because ESS is determined by the target-behavior weights rather than the estimator formula.

The benchmark is included to show that the paper’s conclusion is not based on cherry-picking an estimator that wins every conventional metric. The standard methods trade places across ETFs, while the scientific

question is whether a pre-specified evaluation procedure can support the candidate–baseline decision. The main result is therefore not an estimator ranking; it is the mismatch between the scale of the policy difference and the uncertainty of the evidence used to compare it.

Robustness of the Policy-Resolution Finding

The main protocol already uses three ETFs, non-overlapping forward windows, repeated simulated logs, and time-ordered nuisance estimation. We add three compact checks to ask whether the two central findings—the advantage of retaining the full historical evidence and the weak resolution of the candidate–baseline gap—depend on one narrow financial specification.

Transaction costs

The baseline cost is $c = 0.0005$. We also evaluate

$$c \in \{0, 0.0005, 0.0010\}, \quad (15)$$

using the same values for all ETFs.

ETF	Cost	Lowest-MAE evidence	Median gap (bp)	Median SE (bp)
QQQ	0	All history	0.29	4.07
QQQ	0.0005	All history	0.31	3.42
QQQ	0.0010	All history	0.30	3.13
SPY	0	All history	0.26	2.07
SPY	0.0005	All history	0.26	1.87
SPY	0.0010	All history	0.27	2.51
IWM	0	All history	0.33	1.85
IWM	0.0005	All history	0.35	1.86
IWM	0.0010	All history	0.35	2.21

Table 6: Transaction-cost robustness. The same evidence-window ranking and policy-resolution comparison are recomputed at each cost.

Across all nine ETF–cost combinations, all-history evidence remains the lowest-MAE rule. More importantly, the median policy gap remains much smaller than its bootstrap standard error at every cost. The policy-resolution result is therefore not created by the baseline turnover-cost value.

Risk-sensitive reward

We also use one pre-specified risk-sensitive reward,

$$r_{j,t+1}^{(\lambda)}(a) = w_{j,t}^+(a)R_{j,t+1} - c|w_{j,t}^+(a) - w_{j,t}| - \lambda[w_{j,t}^+(a)]^2[\widehat{\sigma}_{j,t}^{(20)}]^2. \quad (16)$$

The final protocol fixes $\lambda = 0.5$ for all ETFs. The scale is set economically rather than from OPE performance: at 1% daily volatility and full exposure, the risk penalty is about 0.5 basis points per day, the same order as the 0.5-basis-point turnover charge created by a ten-percentage-point exposure change under $c = 0.0005$.

Under this reward, all-history evidence again has the lowest MAE for QQQ, SPY, and IWM. The median gap/SE pairs remain small: 0.29 bp versus 3.00 bp for QQQ, 0.25 bp versus 2.33 bp for SPY, and 0.36 bp versus 2.09 bp for IWM. The main decision conclusion therefore survives a risk-sensitive reward: the uncertainty scale remains several times larger than the candidate–baseline difference.

Alternative policy-training contrast

The main target-policy contrast trains the baseline on 756 observations and the candidate on 126. The single pre-specified alternative replaces the candidate window with 252 observations while keeping the baseline at 756 and leaving the remaining protocol unchanged.

The alternative contrast produces Gap/SE ratios of 0.08 for QQQ, 0.14 for SPY, and 0.16 for IWM, with point-selection accuracy of 50.6%, 49.4%, and 48.8%, respectively. The inability to resolve the candidate–baseline ordering is therefore not specific to the 126-observation candidate-training window.

Taken together, these checks support a narrow but stable conclusion: across the tested transaction costs, reward specification, and policy-training contrast, the historical evidence can estimate broad policy values while remaining too noisy to rank the close candidate–baseline pair reliably.

Discussion and Limitations

Interpretation: value estimation is not policy resolution

The results separate three questions that are often compressed into a single judgment about OPE quality. First, how much historical evidence should be retained? In these experiments, shortening the pre-anchor window increases DR error for every ETF, so the gain in temporal proximity does not compensate for the loss of information. Second, does the behavior log cover actions favored by the target policy? Target-aware re-logging increases ESS, showing that overlap can be improved on the same recent contexts, but the corresponding MAE gains are modest. Third, is the remaining uncertainty small enough to distinguish the policies that matter for deployment? For the candidate–baseline comparison, the answer is no: the gap is about 0.3 basis points while paired standard errors are about 1.9–3.4 basis points.

This third question is the most consequential. A method can improve MAE or ESS without changing the operational fact that the policy difference is unresolved. Conversely, a point estimate always produces a ranking even when the data contain little information about the sign of the difference. The near-50% point-selection accuracy observed here is therefore best read as evidence of insufficient resolution, not as evidence that point estimates are an adequate deployment rule.

The one-sided lower-confidence rule makes the consequence explicit. When the evidence does not separate the candidate from the baseline, it mostly retains the baseline rather than converting numerical noise into a policy switch. This abstention is not a claim that the candidate is worse. It is a statement about what the historical log can support. For close policies, the decision-relevant object is the uncertainty of the paired contrast, not the apparent precision of two marginal value estimates viewed separately.

The results also caution against treating effective sample size as a universal score of OPE reliability. ESS describes concentration in importance weights and is useful for diagnosing target–behavior support. It does

not measure temporal transfer and does not directly measure whether the candidate–baseline gap is large relative to uncertainty. Better support can therefore be real and still be insufficient to resolve a close decision.

What the study does and does not establish

The paper provides a controlled empirical demonstration, not a universal impossibility theorem. It shows that, under one frozen cross-ETF protocol, the best-performing historical evidence rule among those tested still leaves a practically close policy comparison unresolved. The result is strongest as a methodological warning: reporting lower OPE error or higher ESS is not enough to establish that a deployment decision is supported. A policy-resolution analysis should report the paired gap, its uncertainty, and the rate at which a conservative rule can make a decision without forcing a ranking.

Limitations

The evaluation benchmark is one-step and conditional on the logged pre-action position, so it does not reconstruct the multi-period trajectory generated by another sequence of allocation decisions. The behavior logs are simulated rather than collected from a production trading system. The action space and reward are deliberately simple, and the price-taking assumption omits market impact, taxes, liquidity constraints, and interactions among assets.

The target policies are data-adaptive: they are fitted from pre-anchor logged observations that can also contribute to the OPE evidence. The scored forward window is time-separated from fitting and OPE construction, but the design should not be interpreted as evaluation of an externally fixed policy trained on an independent sample.

QQQ, SPY, and IWM provide different market paths, but they are not independent draws from a population of financial markets. The common protocol improves comparability across the tested assets and periods, while broader external validity remains open. Accordingly, the results support statements about the three ETF environments studied here rather than a universal rule for nonstationary financial markets.

Conclusion

A value estimate is not yet a policy decision. In the QQQ, SPY, and IWM experiments, retaining more historical evidence materially improves DR value accuracy, and target-aware re-logging improves action-support diagnostics. Neither result implies that the candidate–baseline choice is resolved. Under the best-performing historical evidence rule, the typical policy gap is only 0.26–0.35 basis points while paired bootstrap standard errors are 1.86–3.42 basis points, leaving point-estimate selection near chance and a one-sided confidence rule mostly abstaining. The practical conclusion is therefore sharper than an estimator comparison: historical OPE should distinguish value accuracy, target-policy support, and policy-gap resolution. When the gap is small relative to uncertainty, retaining the baseline is an evidence-based abstention rather than a failure to produce a numerical estimate.

Code and Data Availability

The formal run archives the source code, frozen protocol, derived result tables, figures, package versions, data hashes, and source-code hash used to produce the reported results. Public replication materials for this study will be released through the Sequential Decision Research GitHub organization at <https://github.com/SequentialDecisionResearch>. A study-specific repository link will replace the organization-level link when the replication package is posted. Market prices were obtained from Yahoo Finance through `yfinance` 1.7.0 on August 29, 2026; the acquisition record specifies the adjusted-close convention and sample dates. Raw vendor market data are not redistributed.

AI-Assisted Editing Disclosure

Generative-AI tools were used for language editing and coding assistance. The author verified the analyses, numerical results, citations, and conclusions and takes responsibility for the content of the manuscript.

OPE and Logging Details

Simulated behavior logger

For context $x_{j,t}$, let $p_{j,t}^{\text{TS}}(a \mid x_{j,t})$ be the Monte Carlo Thompson probability that action a is optimal. The simulated logging distribution is

$$\mu_{j,t}(a \mid x_{j,t}) = (1 - K\epsilon)p_{j,t}^{\text{TS}}(a \mid x_{j,t}) + \epsilon, \quad K = 3, \quad (17)$$

with common probability floor $\epsilon = 0.03$. The Thompson probability is approximated with 400 posterior draws at each decision. Each partial-feedback record stores

$$(j, t, x_{j,t}, a_{j,t}, \mu_{j,t}(a_{j,t} \mid x_{j,t}), r_{j,t+1}(a_{j,t})). \quad (18)$$

Only the sampled-action reward is used to update the logger. The full one-step action-reward vector is stored separately and withheld until scoring after the evaluation protocol has been fixed.

Target-policy implementation

Action-specific Ridge reward models use penalty $\alpha = 2$. The baseline model is fitted on the most recent 756 pre-anchor observations and the candidate on the most recent 126. Predicted action rewards are centered and scaled by a robust within-row action-score scale, with lower bound 2.5×10^{-4} , before the softmax mapping. Both policies use the same softmax temperature, 1.50, and the same target-action probability floor, 0.02. The main target-policy contrast is therefore defined by training history rather than by a different policy sharpness parameter.

The target policies are data-adaptive. Their fitting samples come from the pre-anchor logged history and may overlap with observations later used as OPE evidence. The scored forward window is strictly later than both fitting and OPE construction. The paper therefore does not interpret the experiment as evaluation of an

externally trained policy on an independent historical sample.

Standard OPE formulas

Let $\widehat{q}(x, a)$ be the action-specific reward model and $\widehat{m}_i(\pi) = \sum_a \pi(a | x_i) \widehat{q}(x_i, a)$. With evidence weights ω_i and importance ratios $\rho_i(\pi)$, the benchmark estimators are

$$\widehat{V}_{\text{DM}}(\pi) = \frac{\sum_i \omega_i \widehat{m}_i(\pi)}{\sum_i \omega_i}, \quad (19)$$

$$\widehat{V}_{\text{IPS}}(\pi) = \frac{\sum_i \omega_i \rho_i(\pi) r_i}{\sum_i \omega_i}, \quad (20)$$

$$\widehat{V}_{\text{SNIPS}}(\pi) = \frac{\sum_i \omega_i \rho_i(\pi) r_i}{\sum_i \omega_i \rho_i(\pi)}, \quad (21)$$

$$\widehat{V}_{\text{DR}}(\pi) = \frac{\sum_i \omega_i \{ \widehat{m}_i(\pi) + \rho_i(\pi) [r_i - \widehat{q}(x_i, a_i)] \}}{\sum_i \omega_i}. \quad (22)$$

Rolling-origin nuisance predictions and inference

DM and DR use time-ordered nuisance predictions. The first 252 observations form the minimum warm-up. Thereafter, predictions are generated in 63-observation blocks. For a block beginning at index s , the nuisance reward model is fitted only to rows $1, \dots, s-1$ and then applied to that block. The implementation records the last training date for every predicted row and asserts that `training_end` is strictly earlier than `prediction_date`.

The main forward scoring horizon and anchor spacing are both 80 trading observations, so scored windows do not overlap. Candidate–baseline uncertainty is estimated with a paired moving-block bootstrap using 500 resamples and block length 20. The reported decision rule is an approximate bootstrap-based 95% one-sided lower-confidence rule derived from that resampling distribution.

Repeated logs and reproducibility record

The final experiment uses behavior-seed indices $s = 0, \dots, 19$ for each ETF. Random-number streams are generated deterministically from base seed 20260826 together with the ETF symbol and seed index, so the 20 logs per asset can be reproduced without asset-specific tuning. For each OPE estimate, the saved detail data include the ETF, behavior seed, anchor date, target policy, estimator, evidence rule, value estimate, withheld full-action benchmark, absolute and signed error, ESS, and protocol hash. The protocol metadata also record package versions, data hashes, and the source-code hash used for the formal run.

Final Protocol Checklist

Table 7 records the protocol used for the reported scientific run. The same settings are applied to QQQ, SPY, and IWM; the withheld one-step full-action benchmark is used only after these choices are fixed.

Item	Frozen protocol choice
ETFs	QQQ, SPY, IWM
Action set	$\{-1, 0, +1\}$; exposure step $\Delta = 0.10$
Baseline reward	Return minus proportional turnover cost; $c = 0.0005$
Policy-training windows	Baseline 756 observations; candidate 126 observations
Policy temperature/floor	Common temperature 1.50; target-action probability floor 0.02
Target-policy fitting vs OPE evidence	Data-adaptive pre-anchor fitting; fitting rows may overlap OPE evidence, but never the scored forward window
OPE evidence windows	All history; last 252; last 126; last 60
Recency grid	$m \in \{0, 20, 40, 80\}$ recent observations
Target-aware re-logging	Behavior distribution $0.80 \pi_{\text{base}} + 0.20 \pi_{\text{new}}$ on the same recent contexts
Withheld full-action benchmark	Scoring only; never used for method or hyperparameter selection
Forward horizon	$H = 80$ trading observations
Anchor spacing	80 trading observations
Nuisance estimation	252-observation warm-up; 63-observation rolling-origin blocks
Behavior logger	Linear Thompson sampling; 400 Monte Carlo draws; probability floor 0.03
Behavior seeds	20 per ETF; seed indices $0, \dots, 19$ with base seed 20260826 and deterministic asset-specific mapping
Bootstrap/inference	Paired moving-block bootstrap; 500 resamples; block length 20; approximate bootstrap-based 95% one-sided lower-confidence rule
Risk-sensitive λ	0.50
Alternative policy-training contrast	Baseline 756 observations; candidate 252 observations

Table 7: Final frozen protocol used in the reported cross-ETF experiments.

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